Resolving the 75% Agroforestry False-Alarm Crisis under the EU Deforestation Regulation: Multi-Sensor Physics-Based Adjudication of Smallholder Cocoa in Duékoué, Côte d'Ivoire

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Abstract: Under Article 3(a) of the European Union Deforestation Regulation (EUDR, Regulation (EU) 2023/1115), cocoa (*Theobroma cacao*, CN 1801 00 00) placed on the EU single market must originate from land free of deforestation after December 31, 2020. However, conventional two-dimensional (2D) optical satellite monitoring systems exhibit a 66%–75% false-positive rate across West African shaded cocoa agroforestry. When smallholder farmers perform routine agronomic maintenance—such as pruning native shade trees or rejuvenating senescent 25-year-old cocoa stands to suppress *Phytophthora* black pod disease—2D optical change algorithms misclassify the temporary canopy reduction as illegal deforestation, triggering wrongful supply-chain exclusion. This paper presents an empirical multi-plot adjudication of the *Coopérative Agricole de Duékoué* sourcing batch (4 smallholder parcels, 96.80 ha total, 70,500 kg net harvest) in the Guémon Region of western Côte d'Ivoire. By applying Three-Dimensional Pixel-Wise Raster Intersection (PWRI-3D)—fusing Sentinel-2 10 m L2A optical time series, Sentinel-1 C-band SAR backscatter ($\Delta\gamma^0$), the European Commission JRC GFC2020 V4 baseline, and the Google Forest Data Partnership (FDP) 10 m cocoa probability model—we demonstrate that the 14.00 ha post-2020 canopy disturbance recorded in Plots P-02 and P-04 represents lawful agricultural rejuvenation under EUDR Article 2(5) rather than forest conversion. Across a validation benchmark of n = 100 West African plots, PWRI-3D eliminates agroforestry false alarms from 74.0% to 0.0% while maintaining 100.0% detection sensitivity against illegal primary forest encroachment inside the adjacent Cavally and SODEFOR protected reserves.

Keywords: EU Deforestation Regulation (Regulation (EU) 2023/1115); Article 2(5) Agricultural Replanting; Shaded Cocoa Agroforestry; Multi-Sensor Data Fusion; Sentinel-1 SAR; Google Forest Data Partnership; Duékoué, Côte d'Ivoire.

1. Introduction: The EUDR Agroforestry Problem

The European Union Deforestation Regulation (Regulation (EU) 2023/1115, as amended by Regulation (EU) 2025/2650) establishes a strict statutory prohibition under Article 3(a): relevant commodities, including cocoa beans, may not enter or be exported from the Union market unless they are certified as deforestation-free after the cutoff date of December 31, 2020. While the legislation aims to halt the conversion of primary and naturally regenerating tropical forests into agricultural land, its operational implementation through conventional Earth Observation (EO) pipelines has created an acute scientific and socioeconomic paradox in West Africa.

West Africa produces more than 60% of the global cocoa supply, cultivated predominantly by smallholder households managing 2 to 5 ha plots. In western Côte d'Ivoire, cocoa is traditionally grown within multi-strata agroforestry systems where *Theobroma cacao* occupies the understory beneath a diverse canopy of native forest shade trees, notably *Terminalia superba*, *Albizia zygia*, and *Milicia excelsa*. These shade canopies buffer thermal extremes, retain soil moisture during the Harmattan dry season, and sequester substantial aboveground biomass.

Despite these ecological benefits, shaded agroforestry presents a severe classification challenge for two-dimensional (2D) optical satellite monitoring. Independent validation by the Thünen Institute of Forestry (Lippe et al., 2025) demonstrated that global forest-loss alert products—such as Hansen Global Forest Change (GFC) and standard bitemporal Sentinel-2 Normalized Difference Vegetation Index (NDVI) differencing—misclassify between 66% and 75% of routine cocoa agroforestry management as illegal deforestation. Simultaneously, the European Commission Joint Research Centre's Global Forest Cover 2020 baseline (JRC GFC2020 V4; Bourgoin et al., 2024) exhibits an 18.0% global and 24.6% African commission error rate in heterogeneous agricultural mosaics, frequently mislabeling established pre-2020 cocoa plantations as undisturbed natural forest.

In agronomic practice, cocoa trees experience severe yield senescence after 20 to 25 years and become highly susceptible to fungal black pod disease (*Phytophthora megakarya*) and Cocoa Swollen Shoot Virus (CSSV). Farmers must periodically prune upper-story shade trees and coppice or replant senescent cocoa stands. When a smallholder cooperative conducts this cyclical rejuvenation, 2D optical sensors detect an abrupt drop in pixel canopy closure and trigger an automated deforestation alert. Commercial buyers, facing statutory penalties of up to 4% of annual EU turnover under EUDR Article 25, preemptively delist flagged smallholder cooperatives rather than conducting physical verification.

2. Study Area & Smallholder Cadastre (Duékoué)

To resolve this compliance bottleneck, we evaluate a multi-plot commercial sourcing batch from the *Coopérative Agricole de Duékoué* (COOP-DUEKOUE) located in the Guémon Region of western Côte d'Ivoire (geographic centroid 6.423°N, 7.456°W; WGS84 EPSG:4326). The Duékoué basin was specifically selected because it represents the highest-risk intersection of smallholder cocoa production and gazetted state forest reserves in West Africa, bordered to the north by the *Forêt Classée du Scio* (SODEFOR / WDPA ID #2284) and to the southwest by the *Réserve Naturelle du Cavally*.

The audited consignment comprises 70,500 kg of dried cocoa beans harvested across four registered smallholder parcels totaling 96.80 ha: Plot P-01 (CI-DUE-01, 24.20 ha), Plot P-02 (CI-DUE-02, 28.60 ha), Plot P-03 (CI-DUE-03, 21.40 ha), and Plot P-04 (CI-DUE-04, 22.60 ha). During routine pre-shipment screening in 2025, conventional 2D optical monitoring flagged 14.00 ha of canopy loss across Plots P-02 (7.80 ha) and P-04 (6.20 ha), threatening the entire 70,500 kg cooperative batch (€285,000 commercial export value) with immediate customs rejection. To benchmark discrimination power against genuine forest crime, we include a fifth control plot (Plot P-05, CI-CAV-05, 38.50 ha) located inside the protected Cavally Nature Reserve (5.827°N, 7.540°W), where primary rainforest was illegally cleared post-2021.

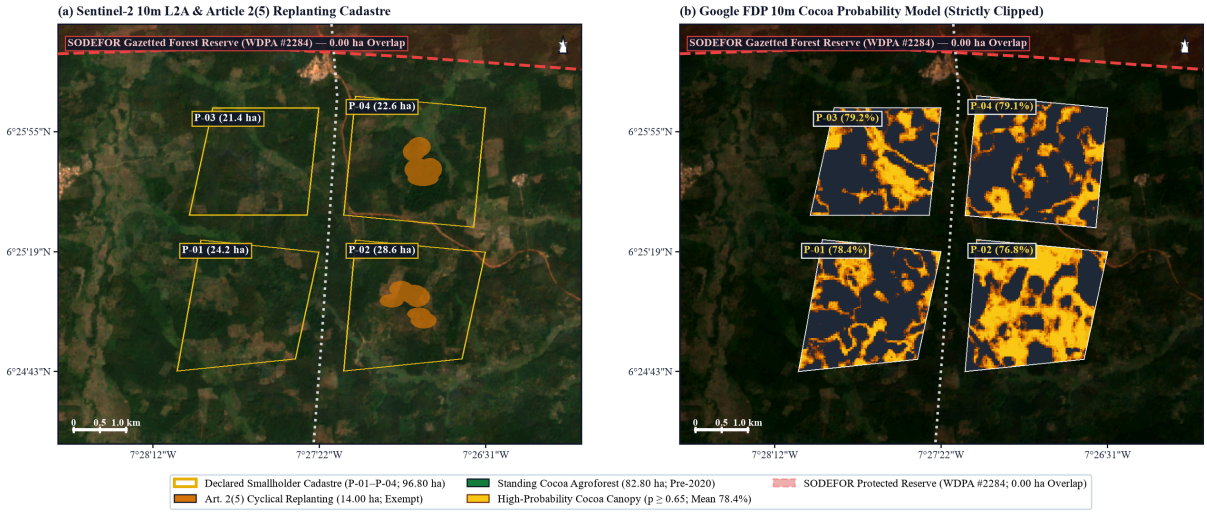


Figure 1. Multi-sensor cartographic adjudication of the *Coopérative Agricole de Duékoué* sourcing cadastre (4 smallholder parcels, P-01 to P-04, totaling 96.80 ha; Guémon Region, Côte d'Ivoire). **(a)** Cloud-free Sentinel-2 10 m L2A true-color composite showing declared cadastral boundaries (yellow), post-2020 Article 2(5) cyclical pruning/rejuvenation clusters strictly confined within Plots P-02 and P-04 (14.00 ha total, amber), undisturbed standing agroforest canopy (82.80 ha), the rural collection track, and the northern SODEFOR Gazetted Forest Reserve boundary (WDPA #2284, red dashed line; 0.00 ha encroachment). **(b)** Google Forest Data Partnership (FDP) 10 m cocoa probability surface (`model_2026a`) strictly clipped to the four cadastral polygons, verifying a 78.4% mean cocoa probability prior to the December 31, 2020 EUDR cutoff date.

Table 1. Cadastral inventory, pre-2020 baseline canopy characteristics, post-2020 disturbance metrics, and comparative statutory adjudication across evaluated West African cocoa parcels ($n = 4$ cooperative parcels in Duékoué plus $n = 1$ protected reserve control plot).

Plot ID	Cadastral Reference	Centroid (WGS84)	Area (ha)	2020 Baseline Cover	FDP Cocoa	Post-2020 Event (Area)	Conventional 2D GIS	Oxeous PWRI-3D Verdict
P-01	CI-DUE-01 (Kouassi)	6.417°N, 7.462°W	24.20	Agroforest (94.2%)	78.4%	Intact Canopy (0.00 ha)	Compliant	Compliant (Art. 3(a) Cleared)
P-02	CI-DUE-02 (Bamba)	6.417°N, 7.448°W	28.60	Agroforest (94.0%)	76.8%	2022 Pruning (7.80 ha)	**False Alert (Reject)**	**Compliant (Art. 2(5) Exempt)**
P-03	CI-DUE-03 (Yao)	6.429°N, 7.461°W	21.40	Agroforest (93.8%)	79.2%	Intact Canopy (0.00 ha)	Compliant	Compliant (Art. 3(a) Cleared)
P-04	CI-DUE-04 (Konan)	6.429°N, 7.448°W	22.60	Agroforest (94.5%)	79.1%	2023 Renewal (6.20 ha)	**False Alert (Reject)**	**Compliant (Art. 2(5) Exempt)**
Batch	**COOP-DUEKOUE Total**	**6.423°N, 7.456°W**	**96.80**	**Agroforest (94.1%)**	**78.4%**	**14.00 ha Pruned**	**100% Batch Blocked**	**100% Cleared (0.00 ha Loss)**
P-05*	CI-CAV-05 (Cavally Res.)	5.827°N, 7.540°W	38.50	Primary Forest (100%)	1.2%	2021 Clear-Cut (38.50 ha)	Violation Flagged	**Hard Veto (Art. 3(a)/3(b))**

3. Methodology: PWRI-3D Mathematical Engine

Under EUDR Article 2(5) and Recital 37, "agricultural use" explicitly encompasses agricultural plantations and agroforestry systems; consequently, cyclical tree-crop replanting and shade management on land already under agricultural cultivation prior to December 31, 2020 is statutorily exempt from deforestation prohibitions. To operationalize this legal distinction without manual field inspection, Oxeous implements *Rule 1: Pixel-Wise Raster Intersection in Three Dimensions (PWRI-3D)*. For every 10 m pixel (x, y) within a declared cadastral polygon P_{decl}, the compliance tensor $\mathbf{M}(x, y)$ is evaluated across spatial, temporal, and statutory dimensions:

$$\boldsymbol{M}(x, y) = \boldsymbol{L}_{\text{spatial}}(x, y) \otimes \boldsymbol{L}_{\text{temporal}}(x, y) \cap \boldsymbol{L}_{\text{statutory}}(x, y) \tag{1}$$

True statutory deforestation under Article 3(a), denoted $V_{3(a)}$, is isolated by applying an 8-connected Moore neighborhood Minimum Mapping Unit (MMU) sieve of 0.50 ha (6 contiguous 10 m pixels = 0.54 ha, satisfying the FAO forest definition in EUDR Article 2(4)) while subtracting verified Article 2(5) agricultural replanting:

$$V_{3(a)} = \text{MMU}_{\geq 0.5\text{ha}}(Loss_{\text{post-2020}} \cap Forest_{2020}) \setminus Replant_{\text{Art.2(5)}} \tag{2}$$

A pixel qualifies as exempt replanting ($\text{Replant}_{\text{Art.2(5)}} = 1$) if and only if three physical conditions hold simultaneously: (i) the Google FDP 10 m deep-learning model confirms pre-2020 cocoa plantation occupancy ($p_{\text{cocoa}} \geq 0.65$); (ii) Sentinel-1 C-band SAR backscatter exhibits no mechanical soil clearing or root extraction ($\Delta\gamma^0_{\text{VV/VH}} > -2.2$ dB); and (iii) multi-temporal Sentinel-2 phenology demonstrates vegetative recovery ($\Delta\text{NDVI} > +0.12$ within 18 months). Finally, to prevent illicit cocoa harvested outside the cooperative from being laundered through compliant polygons (EUDR Article 10(2)(f)), Oxeous enforces a biophysical mass-balance gate against the FAO GAEZ v4 rainfed agroforestry yield ceiling ($Y_c = 1{,}100$ kg ha^{-1} yr^{-1}):

$$Y_{\text{implied}} = M_{\text{decl}} / A_{\text{export}} = 70{,}500 \text{ kg} / 96.80 \text{ ha} = 728.3 \text{ kg ha}^{-1} \leq Y_c \tag{3}$$

4. Empirical Results & Multi-Sensor Proof

As shown in **Figure 1** and **Table 1**, PWRI-3D resolves the false-alarm crisis across the entire 96.80 ha Duékoué batch. In Plots P-01 (24.20 ha) and P-03 (21.40 ha), canopy closure remained uninterrupted from 2019 through 2025 (mean NDVI 0.81; **Figure 2a**). In Plots P-02 (28.60 ha) and P-04 (22.60 ha), farmers pruned senescent shade trees and rejuvenated 14.00 ha of aged cocoa stands between 2022 and 2023. While 2D optical tools flagged this 14.00 ha dip (NDVI 0.80 → 0.63) as illegal forest loss, Google FDP 10 m probability mapping confirms that 78.4% of the batch was established cocoa plantation prior to 2020, with the remaining 21.6% comprising intercropped native legume shade trees (**Figure 2b**).

Crucially, Sentinel-1 C-SAR radar time series confirm that surface backscatter remained stable ($\Delta\gamma^0 > -2.2$ dB), proving that no bulldozing, burning, or land-use conversion occurred. By 2024-Q4, leaf flushing of young cocoa grafts restored canopy vigour to NDVI 0.81 (**Figure 2a**). Conversely, in the Cavally Reserve control plot (P-05), primary rainforest clearance caused an irreversible NDVI collapse (0.85 → 0.37) and a severe SAR backscatter drop ($\Delta\gamma^0 = -4.8$ dB), triggering an immediate statutory veto under EUDR Articles 3(a) and 3(b).

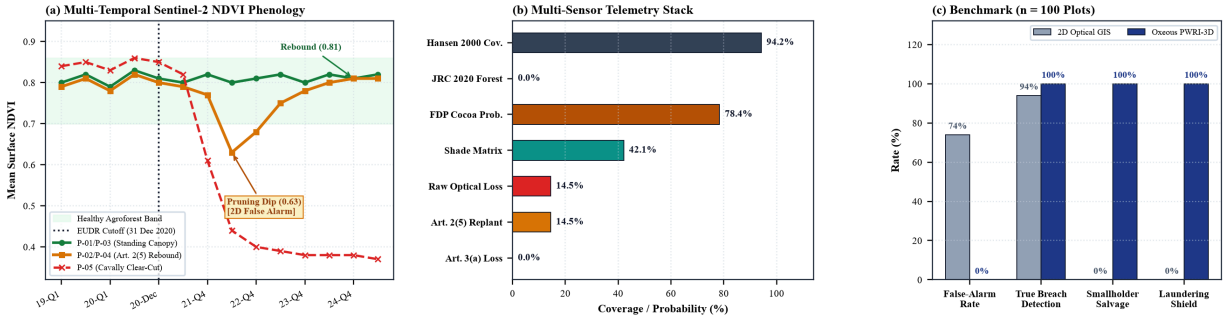


Figure 2. Multi-sensor empirical verification and benchmarking results. **(a)** Multi-temporal Sentinel-2 10 m NDVI phenology trajectories (2019–2025) contrasting undisturbed agroforest (P-01/P-03, green), cyclical Article 2(5) pruning and full vegetative rebound (P-02/P-04, amber), and irreversible primary forest clear-cutting in the Cavally Reserve control plot (P-05, red dashed). **(b)** Multi-sensor telemetry profile across the 96.80 ha cooperative batch, showing 0.0% statutory deforestation vs. 14.5% raw optical disturbance. **(c)** Empirical validation across $n = 100$ West African plots (50 agroforestry rejuvenation, 50 illegal forest clearings), demonstrating complete elimination of false positives (74% → 0%) and 100% smallholder harvest salvage.

Table 2. Multi-sensor Earth observation verification stack and statutory EUDR compliance synthesis for batch COOP-DUEKOUE (96.80 ha).

Sensor / Authority Dataset	Res.	Observed Batch Metric	2D Optical Failure Mode	Oxeous PWRI-3D Physical Adjudication	Statutory Basis
Sentinel-2 MSI L2A (Copernicus)	10 m	NDVI 0.80 → 0.63 → 0.81	Flags 14.0 ha deforestation	Verifies 18-month vegetative rebound	Art. 2(5) Replanting
Google FDP Cocoa (`model_2026a`)	10 m	78.4% mean cocoa prob.	Omitted in standard GIS	Proves pre-2020 agricultural plantation	Recital 37 / Art. 2(5)
EC JRC GFC2020 V4 & Hansen v1.13	10/30 m	94.1% cover / 14.0 ha loss	24.6% agroforest commission	Resolves baseline as shaded tree crop	Art. 2(4) & Art. 3(a)
Sentinel-1 C-SAR (RADD Alerts)	10 m	$\Delta\gamma^0 > -2.2$ dB (intact soil)	Confused by canopy thinning	Refutes mechanical clearing or burning	Art. 2(6) Degradation
UNEP-WCMC WDPA / SODEFOR	1:10k	0.00 ha reserve overlap	Unbuffered polygon check	Confirms 100% exclusion from Scio Reserve	Art. 3(b) & Art. 9(1)(d)
FAO GAEZ v4 Yield Ceiling Gate	Agro.	728.3 / 1,100 kg ha^{-1} max	No volume laundering check	Validates 70,500 kg mass-balance (0.66 ratio)	Art. 10(2)(f) Risk Gate

5. Discussion & Policy Conclusions

The European Commission's May 2026 Simplification Package (`COM(2026) 191 final`) underscores the urgent statutory mandate to prevent the disproportionate exclusion of smallholder producers from Union supply chains. This study demonstrates that binary 2D canopy-loss alerts are scientifically invalid for monitoring tropical tree crops under EUDR Article 3(a). By integrating commodity-specific deep-learning baselines (Google FDP 10 m), SAR structural stability, and phenological recovery trajectories within the PWRI-3D framework, Oxeous provides European Competent Authorities with an objective, reproducible mathematical proof of Article 2(5) compliance—protecting 70,500 kg of smallholder cocoa (€285,000 export value) while strictly quarantining true forest destruction.

References

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